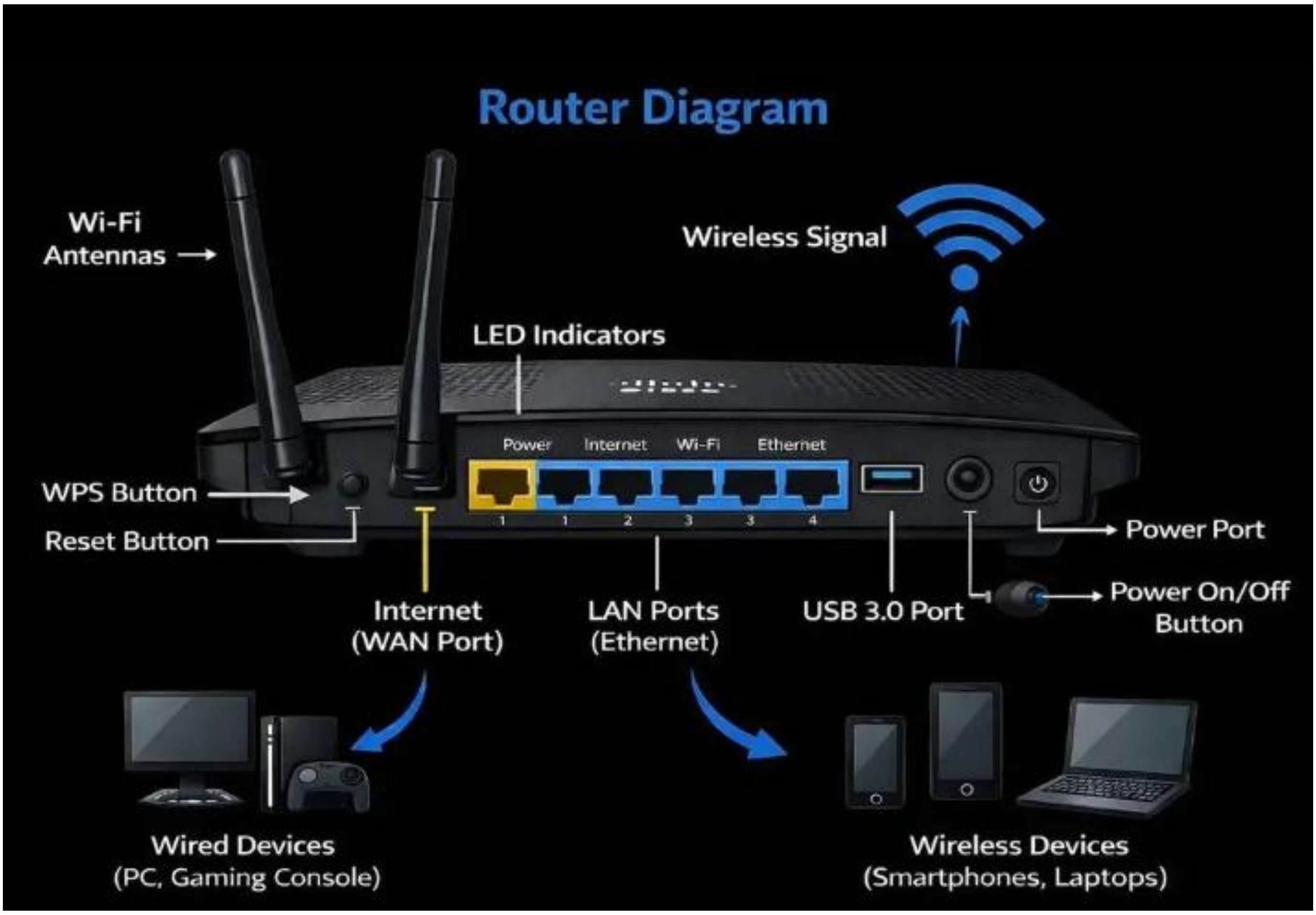
Day -2 : Router and Switch Theory

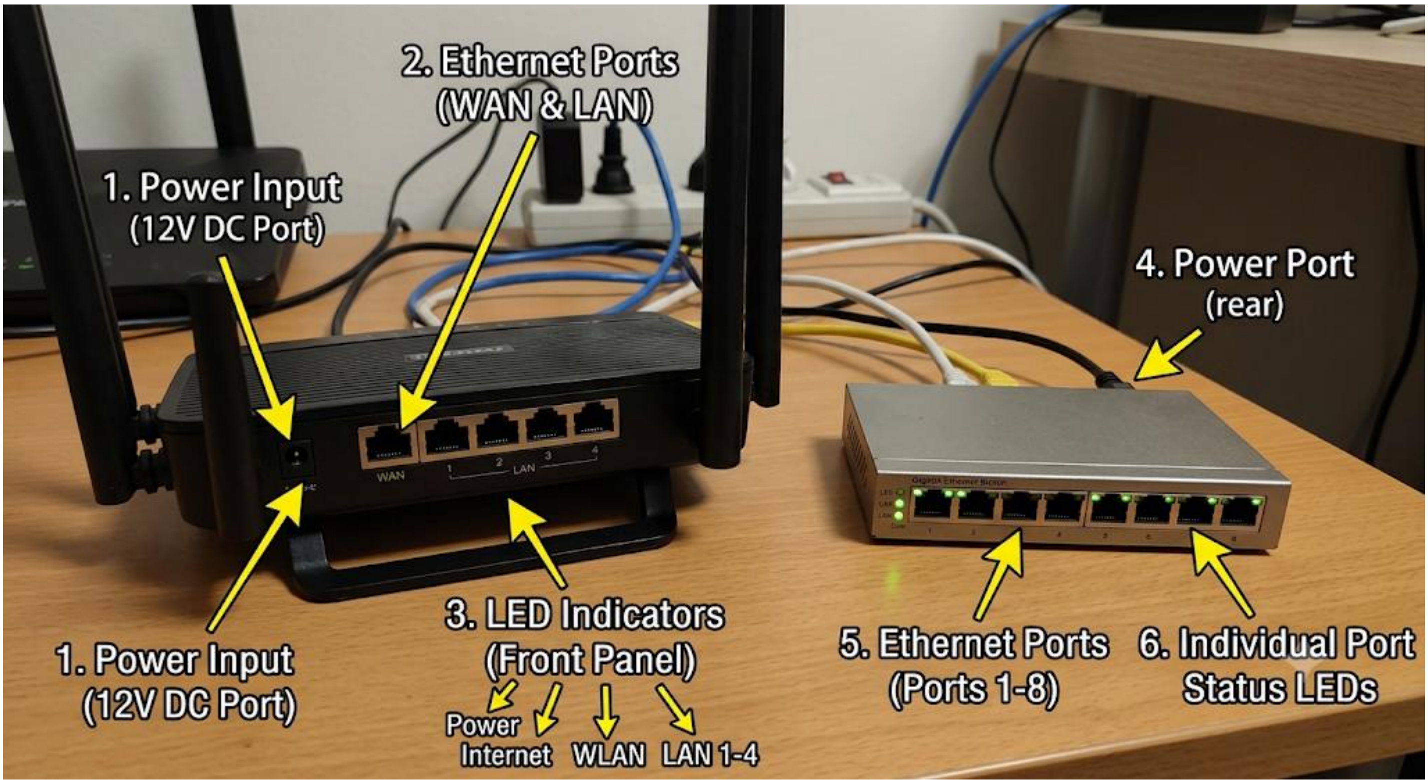
1. Image of wifi router showing WAN,LAN



2. Key Differences Between Routers and Switches

Feature / Aspect	Router	Switch
Primary Function	Connects **different networks** together (e.g., connects your local home/office network to the internet).	Connects **multiple devices together within the same network** (e.g., connecting PCs, printers, and consoles in one room).
Addressing Used	Uses **IP addresses** (Layer 3 - Network Layer) to route data packets to their destination across networks.	Uses **MAC addresses** (Layer 2 - Data Link Layer) to deliver data frames directly to the specific device in the local network.
Traffic Handling	Acts as a **smart traffic director**: reads IP addresses, chooses the best path, assigns IP addresses (via DHCP), and blocks unauthorized external traffic (Firewall).	Acts as a **local dispatcher**: reads MAC addresses and sends incoming data directly to the specific port where the destination device is plugged

3 Image of Router and Switch



4. **Gaming Café Network Setup**

1. Which Device Connects First?

The Router connects first to the ISP (Internet Service Provider) line.

You plug the incoming fiber/internet cable directly into the **WAN (Internet) port** on the router.

2. Connection Steps for the 8 Gaming PCs

[ISP Line] ---> [Router] ---> [Gigabit Switch] ---> [8 Gaming PCs]

1. **Step 1:** Plug the ISP internet cable into the Router's **WAN port**.
2. **Step 2:** Run an Ethernet cable from any **LAN port** on the router to **Port 1** on an 8-port (or 16-port) Gigabit Ethernet Switch.
3. **Step 3:** Plug Ethernet cables from the remaining ports on the switch directly into each of the **8 gaming PCs**.